

THE MAGIC OF LENSES

Optics Lesson & Activities • Grade 4 Science

Welcome, Young Scientists! Today we are going to explore how light can play tricks on our eyes, bend around corners, and make things look giant or tiny! Have you ever wondered how glasses help people see better, or how a tiny smartphone camera captures beautiful pictures? Let's find out!

1. What is a Lens?

A **lens** is a clear, curved piece of glass or plastic that has a very special job: it bends light passing through it! By bending light, lenses can help us see things much bigger or much smaller.

Key Word	What it Means
Lens	A clear, curved piece of glass or plastic that shapes light.
Transparent	An object that is perfectly clear and lets all light pass through it seamlessly.
Refraction	The scientific word for when light bends as it moves from one thing to another (like from air into water or glass).

Try This Experiment at Home!

Take a glass of lemon water and place a straw inside it. Look at the straw from the side. Does it look broken or bent right where it enters the water? It isn't actually broken—that's just **refraction** playing tricks on your eyes because light travels differently through water than air!

2. Meet the Two Main Types of Lenses

Lenses come in different shapes depending on what we want them to do with light. Let's explore the two absolute favorites:



Convex Lens

The Magnifier

A convex lens is thicker in the middle and thinner at the edges. Think of a round, sweet **laddoo shape**!

Memory Trick:

"Convex Comes OUT,
Convex Converges,
Convex = BIG!"

How it works: It bends incoming light rays inward so they meet at a single point (they **converge**). This makes objects look much **BIGGER**!

Real-world uses:

- Magnifying glasses
- School microscopes
- Mobile phone camera lenses



Concave Lens

The Shrinker

A concave lens is thinner in the middle and thicker at the outer edges. Think of a deep **cooking bowl or a karahi shape**!

Memory Trick:

"Concave CAVES IN
like a deep bowl,
Concave = SMALLER!"

How it works: It bends incoming light rays outward, spreading them apart (they **diverge**). This makes objects look much **SMALLER**!

Real-world uses:

- Glasses to fix Myopia (nearsightedness)
- Door peepholes (magic eye) to see who is outside

3. Check Your Progress! (Quick Quiz)

Let's see how much of a lens detective you have become! Circle or check the correct answer below.

Question 1: Which lens is shaped like a delicious laddoo (thick in the middle)?

A) Concave Lens

B) Convex Lens

Question 2: When light passes through a door peephole, everything looks smaller. What lens is used?

A) Concave Lens

B) Convex Lens

Question 3: What do we call it when light plays tricks and bends as it moves through glass or water?

A) Reflection

B) Refraction



Lens Hunt Homework Assignment

Become a real science explorer at home! Find **two objects** around your house that contain a lens (ask your parents or family to help you search!). Draw them and write which lens type they use.

Object #1: _____

Draw it in the box below:

A large, empty rectangular box with a dashed border, intended for drawing the first object.

This object uses a: ☐ **Convex Lens** ☐ **Concave Lens**

Object #2: _____

Draw it in the box below:

A large, empty rectangular box with a dashed border, intended for drawing the second object.

This object uses a: ☐ **Convex Lens** ☐ **Concave Lens**

🌟 You are an amazing Lens Detective! Keep exploring! 🌟